

Chest X-Ray Pneumonia Detection Using Convolutional Neural Networks, Support Vector Machines, and Vision Transformers

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Abstract

This project investigates the classification of chest X-ray images into two categories: *NORMAL* and *PNEUMONIA*. I develop, train, and compare three machine learning models: a Convolutional Neural Network (CNN), a Support Vector Machine (SVM) trained on CNN-extracted features, and a Vision Transformer (ViT). The dataset is merged, balanced, and re-split into stratified train/validation/test sets using a 70–15–15 ratio while preserving a 50/50 class distribution because the dataset was initially imbalanced. Extensive experiments demonstrate that both CNN and ViT achieve strong performance above 93% test accuracy, while SVM performs slightly worse but remains competitive. Confusion matrices and detailed classification reports are included.

1 Introduction

Despite many advances in healthcare over the years, pneumonia remains a prevalent source of morbidity and mortality around the world. Early diagnosis of the disease is very important, since the risk of complications, hospital stay, and mortality can significantly be reduced if the disease is treated in a timely manner. Chest X-ray remains the foremost used technique for the diagnosis of pneumonia.

Despite the above challenges, the radiographic diagnosis of chest X-ray images remains a daunting task even for professional radiologists. The presentation of pneumonia may co-occur with other chest conditions in some patients. Moreover, minute radiographic details may not be easily identifiable. These factors may lead to human error in diagnosis. Additionally, the general public may not be in a position to correctly interpret radiographic images.

This project explores three classification approaches that can help to handle this issues and classify x-rays correctly, I implemented these models:

- A Convolutional Neural Network (CNN) trained end-to-end.
- A Support Vector Machine (SVM) trained on features extracted from the CNN’s convolutional layers.
- A Vision Transformer (ViT) fine-tuned on the dataset.

All models are trained on a balanced dataset, with the goal of evaluating their comparative performance in a controlled setting.

2 Dataset Preparation

2.1 Original Dataset

The dataset consists of chest X-ray images labeled as *NORMAL* or *PNEUMONIA*. The original dataset came pre-split into *train*, *val*, and *test* folders but showed severe imbalance:

- Train: 5216 images (74% PNEUMONIA)
- Val: 16 images (50/50 split but too small)
- Test: 624 images (62% PNEUMONIA)

Such imbalance and improper sizing make the dataset unsuitable for a fair comparison.

2.2 Merging and Class Balancing

All three splits were merged into a single dataset pool. Then:

1. All *NORMAL* samples were kept.
2. *PNEUMONIA* samples were randomly undersampled to match the number of *NORMAL* samples.

This produced a perfectly balanced 50/50 dataset. However, with this approach, we lost almost half of our data because initially, the numbers of the samples were 4273 (73%) PNEUMONIA and 1,583 (27%) NORMAL class data. to equalize the numbers, I dropped 46% of the initial data. But still, for each class, there are 1583 samples, which is an appropriate number for image classification, and by this, I balanced my dataset, which is better for the models.

2.3 Stratified Train–Validation–Test Split

The balanced dataset was split using:

70% training, 15% validation, 15% testing

A stratified splitter ensured that each subset preserved the 50/50 class distribution.

```

Train: 2216
Val: 475
Test: 475

Train label ratio:
labels
PNEUMONIA    0.5
NORMAL       0.5
Name: proportion, dtype: float64

Val label ratio:
labels
PNEUMONIA    0.501053
NORMAL       0.498947
Name: proportion, dtype: float64

Test label ratio:
labels
NORMAL       0.501053
PNEUMONIA    0.498947
Name: proportion, dtype: float64

```

Figure 1: Final dataset split after balancing and stratification.

3 Methods

3.1 Image Preprocessing

All images were converted to grayscale and resized to 224×224 . The following transforms were used:

- Training: random horizontal flip, random rotation, normalization.
These modifications of images, such as rotation and flipping, help the model train for different inputs and guaranty the absence of errors if such images are tested in real life.
- Validation/Test: resize and normalization only.

For ViT, images were converted to 3-channel RGB to match the expected input.

3.2 Convolutional Neural Network (CNN)

Convolutional Neural Network (CNN) refers to a deep learning model tailored specifically for image processing. CNN networks utilize the spatial characteristics of images to analyze images by learning localized patterns in images via convolutional layers. These layers involve sliding filters over images to activate when they notice relevant image characteristics. Early layers of CNN networks usually involve the detection of simple image patterns. Layers further down in a CNN network may involve the detection of more complex image patterns that relate to image classification.

Within the context of the project, the CNN model comprises a series of convolutional layers followed by batch normalization layers, a nonlinear activation function (ReLU) layer, and a Max Pooling layer. Essentially, the approach allows the model to derive robust features that remain invariant under scales. Finally, the convolutional output feature mappings are flattened to allow the fully connected layers of the classifier to perform the necessary high-level reasoning for the prediction of the class (either NORMAL or PNEUMONIA) while dropout was applied. CNNs prove to be particularly advantageous in the analysis of medical images since they have the capability to learn the characteristics of a disease without having to be manually engineered. In the case of chest X-ray images, characteristics including opacity, density, and the structure of the lungs differing in the images can all be learned by the CNN without having to manually implement them. The CNN in the experiment serves as the baseline end-to-end DL approach in the classification of pneumonia. The CNN consists of:

- Four convolutional blocks: Conv $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ MaxPool.
- A fully connected classifier: 512-unit dense layer with ReLU and dropout.

Training setup:

- Loss: Cross-entropy.
- Optimizer: Adam ($\text{lr} = 10^{-4}$, weight decay 10^{-4}).
- Epochs: 10.
- Batch size: 32.

3.3 SVM on CNN Features

Support Vector Machines (SVMs) are well-known machine learning algorithms commonly used for classification purposes. Unlike algorithms that involve the learning of hierarchical features directly from the image pixels, SVMs involve the processing of fixed-length vectors of features in order to search for the best possible decision boundary that maximally distinguishes the different classes. The Radial Basis Function kernel used in the experiment allows for non-linear boundaries.

Unlike traditional SVM implementations in images, in the current analysis, the SVM model isn't trained on the images. Instead, a Convolutional Neural Network (CNN) extracts the

characteristics from the images. After the last classification head of the CNN model has been eliminated, the end result of the convolutional layers undergoes a Global Pooling step to form a compact vector representing high-level image characteristics. These vectors are later used to train the SVM to classify NORMAL images from the PNEUMONIA images.

Leans on the best practices of both deep learning algorithms and traditional machine learning in terms of:

- The CNN extracts a detailed representation of the image.
- The SVM provides a strong classifier based on the developed features.

SVMs may have a strong generalization capability in the case of a small number of examples or a balanced dataset. They also serve as a reference point for understanding the role of end-to-end learning compared to the use of learned features.

After CNN training, the final convolutional feature maps were globally pooled to 256-dimensional vectors. These features were used to train an RBF-kernel SVM with $C = 1.0$.

3.4 Vision Transformer (ViT)

The Vision Transformer is a modern deep learning architecture that reuses the transformer model, originally designed for natural language processing, for computer vision tasks. Unlike CNNs, which rely on convolutions to extract local spatial features, ViT regards an image as a sequence of fixed-size patches. Each patch is flattened and then embedded into a high-dimensional vector, akin to word representations in NLP models. Positional embeddings are added to retain the spatial structure.

These embeddings in a patch are further passed onto a transformer encoder for processing via multi-head self-attention. Self-attention enables the network to take into consideration the entire image in terms of relationships between patterns in different parts of the picture. This makes the ViT highly efficient when there are long range dependencies.

This involves the application of a pre-trained ViT-B/16 model to the Pneumonia dataset. It replaces the final classification head with a two-class output layer; the model trains for several epochs. The use of pre-trained weights here allows faster convergence and better performance, which is very useful in medical imaging because the datasets are usually quite small.

Among the evaluated models, the test accuracy is highest for ViT, indicating its global self-attention mechanism helps in capturing complex radiographic patterns, which may span large areas of the chest X-ray, thus being suitable for pneumonia detection. We fine-tuned a pretrained ViT-B/16 model:

- The final classification head was replaced with a 2-class linear layer.
- We trained for 5 epochs using Adam with $\text{lr} = 10^{-4}$.

4 Evaluation Metrics

To evaluate the performance of the classification models, we report four standard metrics commonly used in medical image analysis: **precision**, **recall**, **F1-score**, and **support**. These metrics are derived from the entries of the confusion matrix, defined as:

TP = True Positives,

TN = True Negatives,

FP = False Positives,

FN = False Negatives.

4.1 Precision

Precision measures the proportion of predicted positive samples that are actually positive. It quantifies how reliable a model’s positive predictions are.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

A high precision indicates that the model produces few false positives.

4.2 Recall

Recall (also called sensitivity) measures the proportion of actual positive samples that are correctly identified by the model.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

A high recall is critical in medical diagnosis tasks such as pneumonia detection, where false negatives (missed illness cases) can have severe consequences.

4.3 F1-score

The F1-score is the harmonic mean of precision and recall. It provides a single balanced measure that accounts for both false positives and false negatives.

$$\text{F1-score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

This metric is especially useful when dealing with imbalanced datasets.

4.4 Support

Support refers to the number of true instances of each class in the dataset. For a given class c :

$$\text{Support}(c) = \text{Number of samples whose true label is } c$$

Support is not a performance metric itself, but it provides context for interpreting precision, recall, and F1-score.

5 Results

The results of all models demonstrated in general high accuracy for test dataset, which shows that all models are good enough to classify x-rays for identifying the pneumonia. In comparison, SVM performed worse than deep learning models, which makes it less reliable in the sense of image classification.

In the context of a medical diagnosis problem such as the detection of pneumonia, the evaluation criterion of highest importance would be the model’s capacity for correct prediction of the “PNEUMONIA” class. A patient having pneumonia being labeled “normal” would prove harmful since a delay in the onset of treatment would lead to critical health conditions. Below, we analyze these models in more detail.

5.1 CNN Training Performance

Convolutional Neural Networks showed great scores in training and validation set by both increasing the accuracy and decreasing the loss for each next epoch. This can be seen in Figure 2 and 3:

Epoch 1/10	Train Acc: 0.8344	Val Acc: 0.9305
Epoch 2/10	Train Acc: 0.9206	Val Acc: 0.9516
Epoch 3/10	Train Acc: 0.9255	Val Acc: 0.9684
Epoch 4/10	Train Acc: 0.9328	Val Acc: 0.9600
Epoch 5/10	Train Acc: 0.9395	Val Acc: 0.9579
Epoch 6/10	Train Acc: 0.9287	Val Acc: 0.9579
Epoch 7/10	Train Acc: 0.9400	Val Acc: 0.9684
Epoch 8/10	Train Acc: 0.9458	Val Acc: 0.9705
Epoch 9/10	Train Acc: 0.9395	Val Acc: 0.9558
Epoch 10/10	Train Acc: 0.9499	Val Acc: 0.9726

Figure 2: Accuracy update for each epoch in CNN training and validation sets

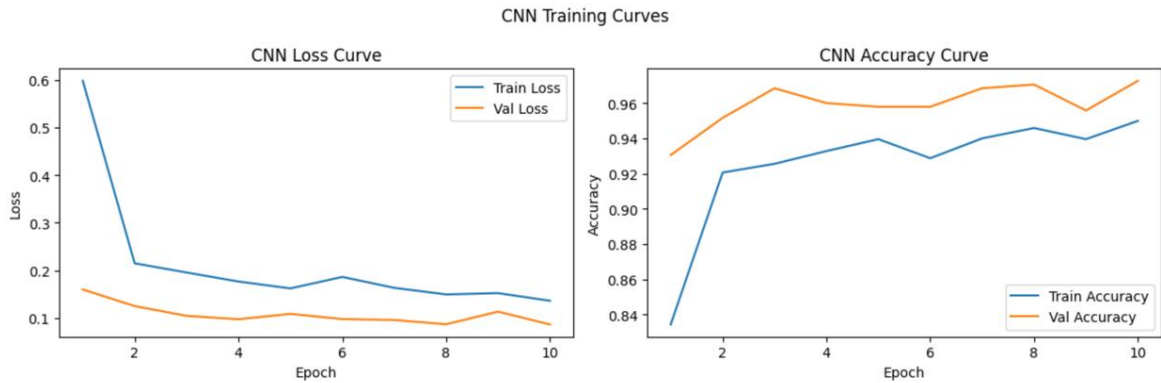


Figure 3: Visual illustration of accuracy and loss change for each epoch in CNN training and validation sets

5.2 CNN Test Results

The most important part of model training is evaluating it on the test dataset. The accuracy of the CNN model for test is over 93%, which is really high score for image classification tasks.

$$\text{CNN Test Accuracy} = 0.9347$$

Table 1: CNN classification report.

Class	Precision	Recall	F1-score	Support
NORMAL	0.95	0.92	0.93	238
PNEUMONIA	0.92	0.95	0.94	237
Accuracy	0.9347			

These three classification schemes have somewhat different characteristics while being compared on the test dataset. As the problem involves the diagnosis of a medical condition, particular attention needs to be given to the metrics for the PNEUMONIA class. In medical diagnosis tasks, a pneumonia case being missed (or a “false negative” prediction) can delay treatment for the patient and may have very serious health repercussions. Hence, the recall value for the pneumonia class is the most important.

The CNN performs well in all aspects. Precision for NORMAL (0.95) is very high. This implies that there are very few cases of healthy people being classified under the sick category. The recall value for the class PNEUMONIA (0.95) is also highly satisfactory. This denotes that

the output model has been able to classify the majority of the images into the correct categories. This is very necessary in a medical application. The F1-scores for the two classes (0.93 and 0.94) validate the result. There is a good balance between the sensitivity and the specificity of the model.

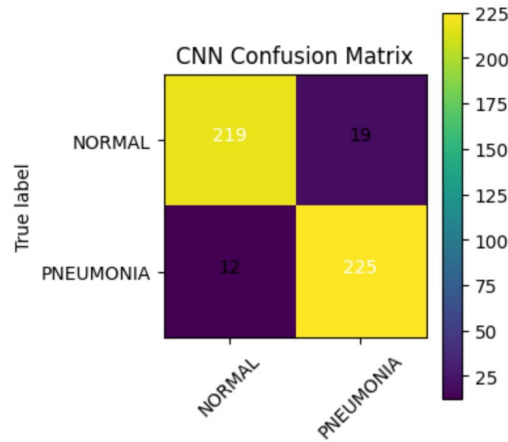


Figure 4: Confusion matrix for CNN

5.3 SVM Results

The SVM holds its ground but clearly stays behind CNN and ViT. For NORMAL cases, its recall is 0.87 - the lowest here, which means more healthy patients are mistaken for having pneumonia. For PNEUMONIA, it is a solid 0.92 yet still not up to CNN or ViT. Precision for PNEUMONIA is 0.87, signaling more false positives. Overall, F1-scores are 0.89 for NORMAL and 0.90 for PNEUMONIA, showing some variability in performance. In summary, the SVM can separate the two classes reasonably well, but in critical medical contexts, it is less trustworthy because it gives more errors than ViT and CNN, notably false negatives.

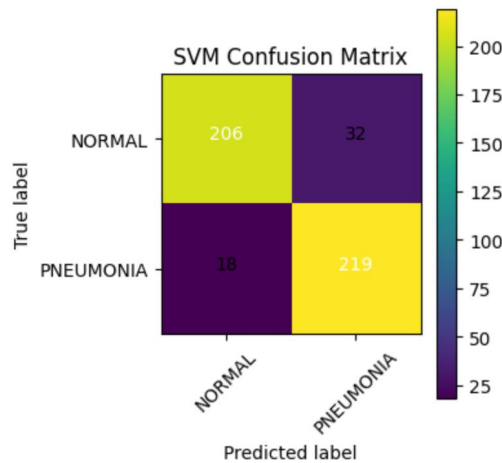


Figure 5: Confusion matrix for SVM

$$\text{SVM Test Accuracy} = 0.8947$$

Table 2: SVM classification report.

Class	Precision	Recall	F1-score	Support
NORMAL	0.92	0.87	0.89	238
PNEUMONIA	0.87	0.92	0.90	237
Accuracy	0.8947			

5.4 ViT Training Performance

In comparison with CNN, the training and validation accuracy for ViT has better scores and hence show better performance, also improving by each epoch. However, the resulting validation set accuracy is greater in CNN, it did not show a big difference, and did not affect the accuracy score for test set, which is shown in the next section.

```
[ViT] Epoch 1/5 | Train Acc 0.8655 | Val Acc 0.9537
[ViT] Epoch 2/5 | Train Acc 0.9567 | Val Acc 0.9642
[ViT] Epoch 3/5 | Train Acc 0.9675 | Val Acc 0.9789
[ViT] Epoch 4/5 | Train Acc 0.9801 | Val Acc 0.9621
[ViT] Epoch 5/5 | Train Acc 0.9716 | Val Acc 0.9663
```

Figure 6: Accuracy change for each epoch in ViT training and validation sets

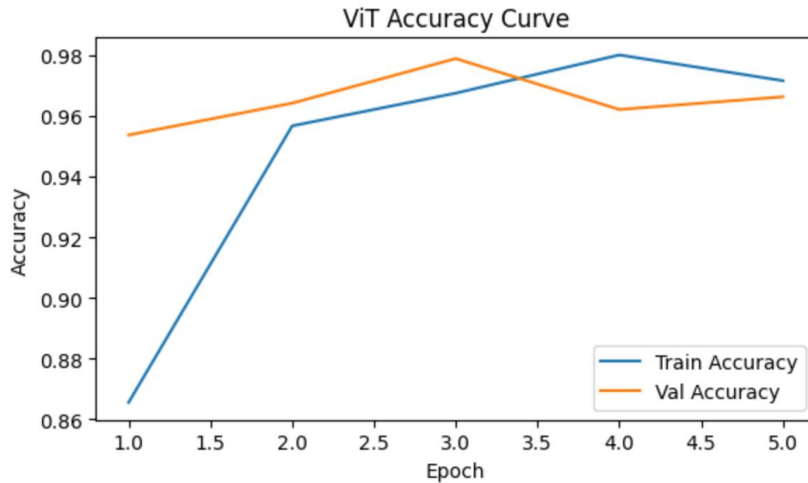


Figure 7: Visual illustration of accuracy change for each epoch in ViT training and validation sets

5.5 ViT Results

The overall performance is led by the Vision Transformer. In the case of NORMAL, it achieves a very impressive precision of 0.96, which is the best of all models and indicates a strong ability in correctly identifying healthy patients. More importantly, its recall for PNEUMONIA is 0.96, the highest sensitivity among the three models, which implies that ViT misses the fewest cases with pneumonia—very crucial for safe clinical use. Both classes achieve an F1-score of 0.94, showing the steady and reliable performance of ViT. This probably results from the knack of ViT for capturing long-range spatial relationships in chest X-ray images.

ViT Test Accuracy = 0.9389

Table 3: ViT classification report.

Class	Precision	Recall	F1-score	Support
NORMAL	0.96	0.92	0.94	238
PNEUMONIA	0.92	0.96	0.94	237
Accuracy	0.9389			

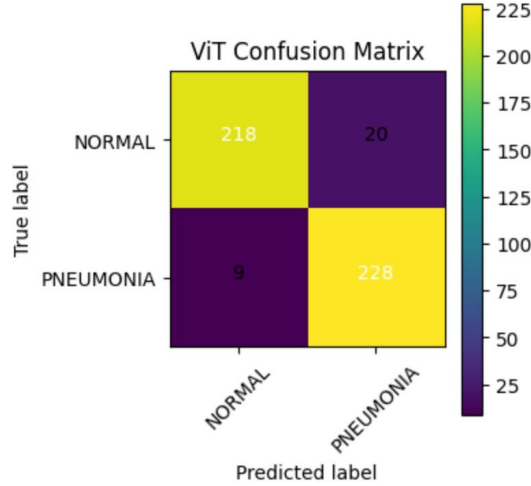


Figure 8: Confusion matrix for ViT

6 Conclusion

By the conducted work we can observe that all models show high scored results but they are not similar. ViT and CNN outperform SVM in almost every key metric. ViT is at the top overall in accuracy and also leads in recall for pneumonia, which means a lot in medical use. CNN follows right behind, delivering solid precision and well-balanced F1 scores, suggesting dependable and consistent classification.

SVM is a good baseline but tends to miss more true cases and flag more non-cases, showing that its feature representation isn't as rich as what end-to-end deep learning models learn. That makes sense: CNNs and ViTs extract hierarchical or attention-guided features directly from the image, while SVM has to rely on predefined embeddings. In particular, the self-attention in the ViT appears to be especially useful for the analysis of chest X-rays, where the model leverages large-scale radiographic patterns with broad lung areas.

In general, the results of this study have demonstrated that deep learning approaches, particularly ViT and CNN, are significantly more suitable for pneumonia detection compared to traditional classifiers. It can be seen from their test set accuracy, which is greater than 93%, whereas SVM results in 89%. Their high sensitivity, strong precision, and well-balanced performance make them more trustworthy for clinical decision support.

Deep learning models, in particular the ViT and CNN, are robust and accurate for automatic pneumonia detection. Future studies could use more substantial datasets with further advanced augmentation, interpretability methods, and external validation to widen the clinical relevance of such models.

References

- Paul Mooney, *Chest X-Ray Pneumonia Dataset*, Kaggle, Available at: <https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>